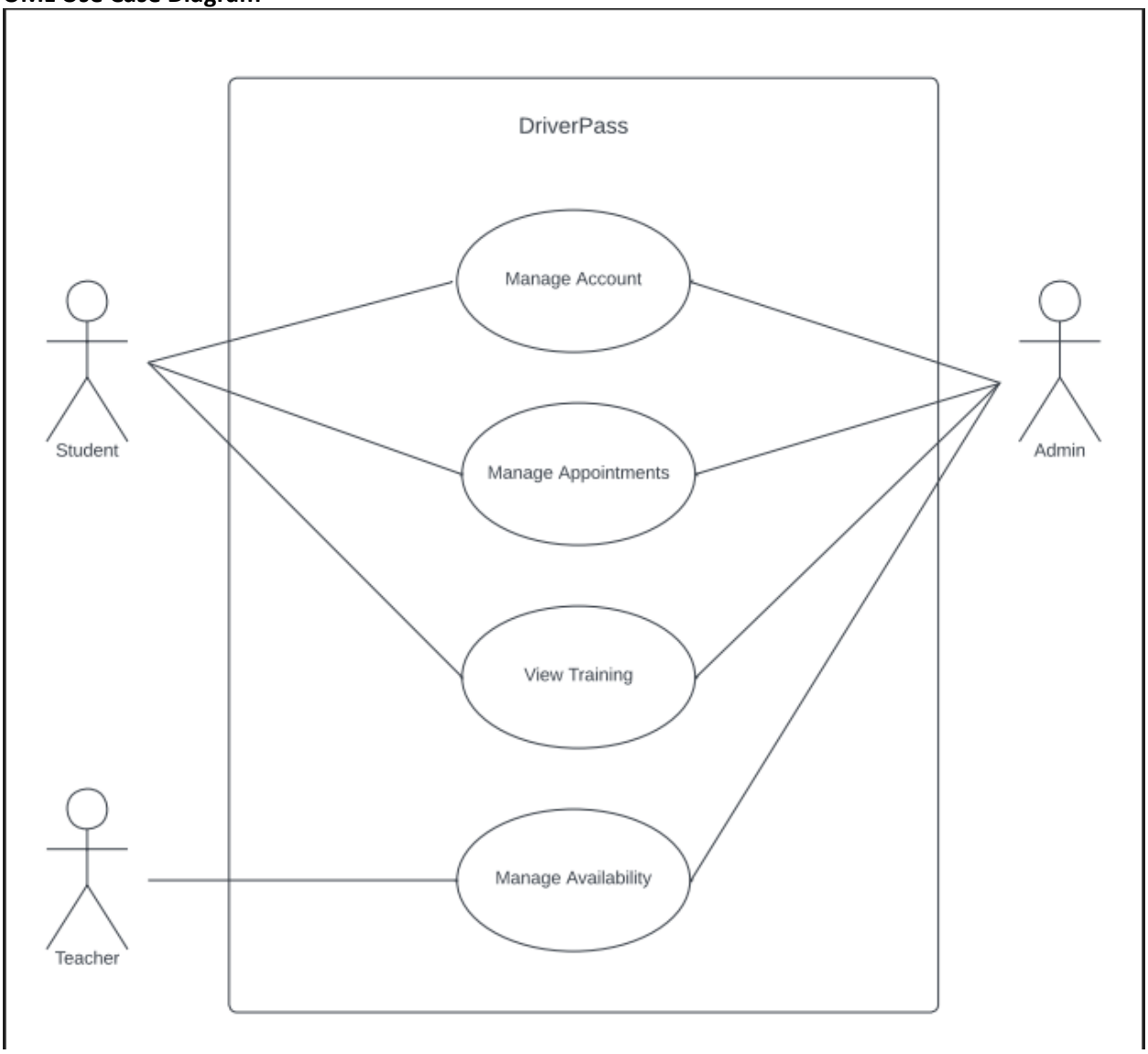
CS 255 System Design Document Template

This template lays out all the different sections that you need to complete for Project Two. Each section has guidance to prompt your thinking. You will need to continually reference the interview transcript as you work to make sure that you are addressing your client's needs. There is no required length for the final document. Instead the goal is to complete each section based on what your client's needs are. Remove this note when you are finished, and replace all bracketed text with the relevant information.

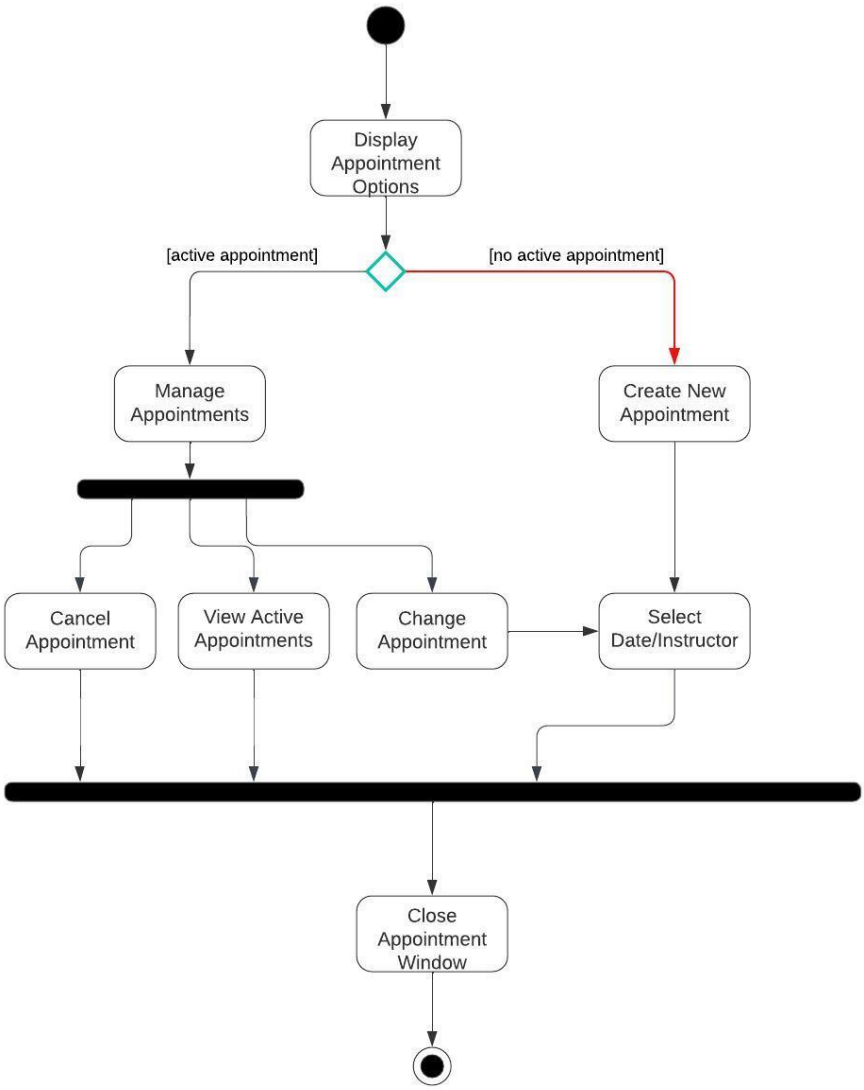
UML Diagrams

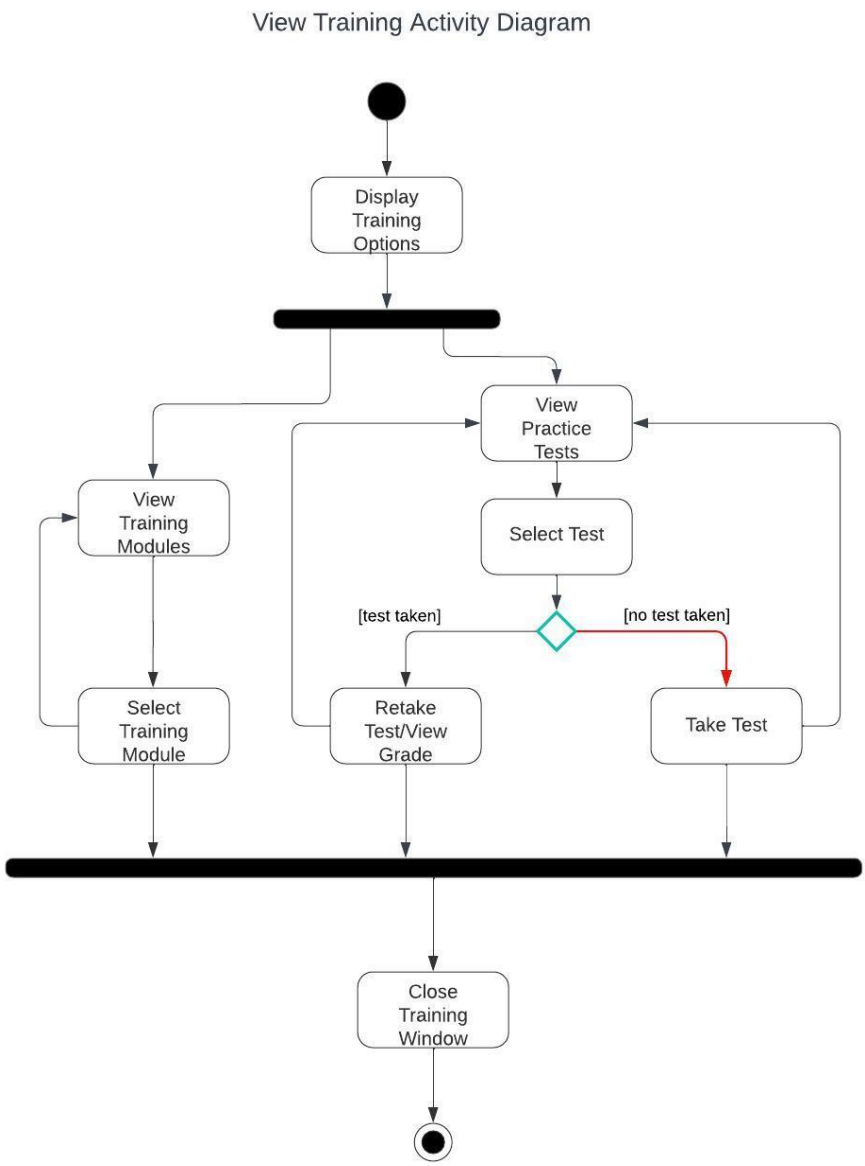
UML Use Case Diagram



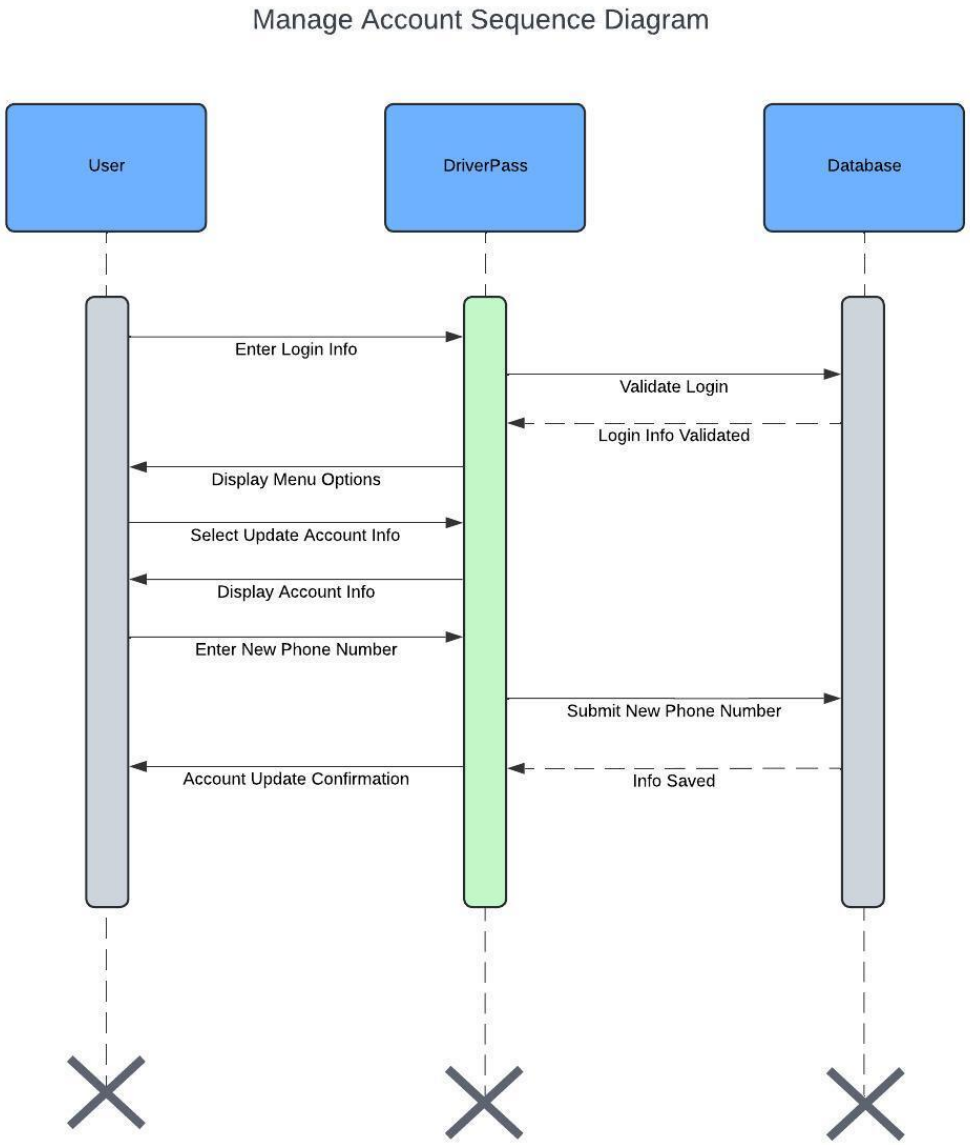
UML Activity Diagrams

Manage Appointments Activity Diagram

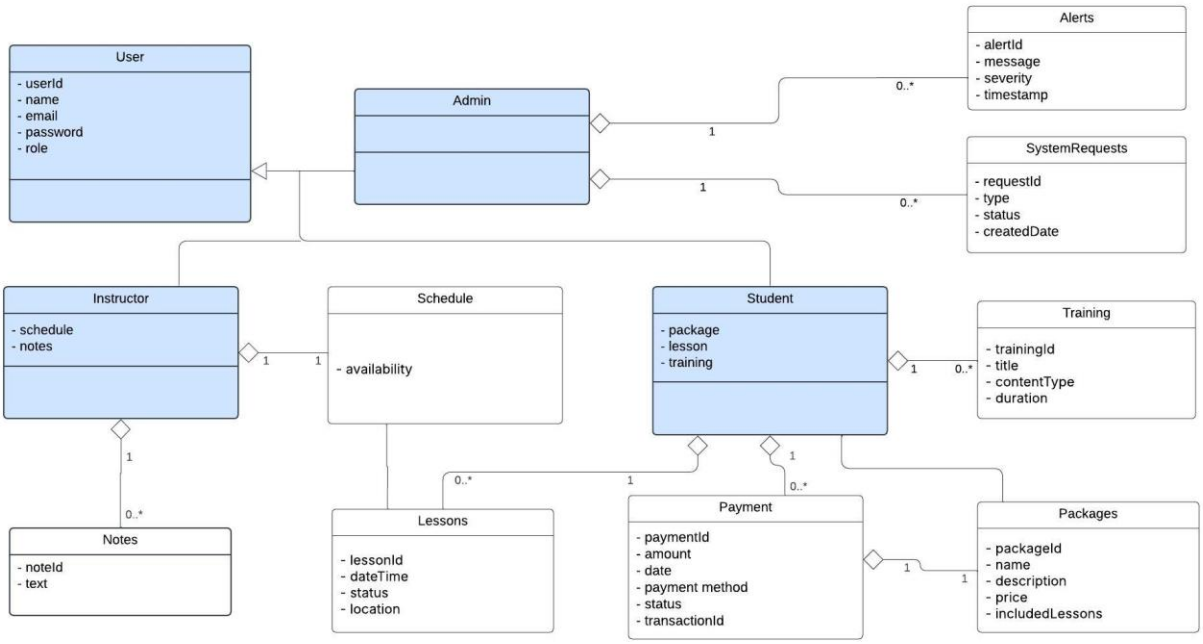




UML Sequence Diagram



UML Class Diagram



Technical Requirements

Hardware Requirements:

- The system needs to be implemented using cloud servers with 8-16 gigabytes of RAM, multicore processors, and SSD storage for quick database access.
- The system needs to support scaling for a potential increase in the userbase.

Software Requirements:

- The system needs to use a database such as MySQL.
- The system will need to run on a secure web server.
- The system also needs to run on Windows, Linux, macOS, iOS, and Android operating systems.
- The system needs to utilize modern front-end technology like HTML5.

Tools:

- The system will be developed using an IDE and will incorporate modern principles of collaborative software development.

Infrastructure:

- The system will be deployed via a cloud platform such as AWS.
- The system will utilize firewall and detection protections.
- The systems will access DMV data.
- Secure authentication for login credentials, and role-based access control are required.
- Account locking mechanisms and suspicious activity monitoring are also required.